

Implementation of the Dynamic Time Warping Algorithm for Bioacustic Detection

by

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* Supervisor X.
* Funding Y.

> *A very long quote*
> *from someone very important.*
> – Mr M

Thank you to Leanne for helping to prepare this template.

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Nomenclature

Acronyms and Abbreviations

MFT My Funky Thesis

Definitions

S

Student is an entity needing a thesis to transcend the state of being a student.

Chapter 1

Introduction

1.1 Background

Since the late 1990s there have been indications that whales have been struck by shipping vessels.[1] Today this has become a larger concern, as vessel speeds and traffic has substantially increased. In False Bay, South Africa, a colony of Bryde's whales have been detected. Bryde's whales are found to migrate in smaller regions instead of the usual large-scale migrations of other whale species. Due to the increase in shipping traffic as well as the fishing activities in the area, the Bryde's whales in this are are also in danger.[2] Detection and classification systems for whale calls can be used to monitor the migration patterns of the specific whale colony throughout the year. The information gained from such analysis can help determine when vessel traffic should be diverted for the safety of the whale population. These types of systems have been implemented on different whale colonies, but the migration patterns of the False Bay colony has not yet been thoroughly documented.[3] Due to the variations in call types between different colonies of some whale species (killer whales[4], blue whales[5]), models trained for other Bryde's whale colonies would not be completely trustworthy for the detection of calls within the False Bay Bryde's Whale colony as universiality of calls cannot be assumed. Therefore, a model trained specifically for the False Bay colony would be beneficial. Dynamic Time Warping (DTW) has been shown to be an effective whale classification method on different whale colonies.[6]

1.2 Problem Statement

This project will determine whether using Dynamic Time Warping (DTW) in the frequency domain, thus using vector features, is able to effectively detect Bryde's Whale calls within the False Bay region. Three feature extraction methods will be compared. These will be

Short Time Fourier Transform (STFT), Mel-frequency Cepstral Coeffficients (MFCC) and Linear Predictive Coding (LPC).

1.3 Objectives

1. Implement a DTW detection system using three different feature extraction methods(STFT, MFCC and LPC). This shows whether the effectiveness of the DTW system is independent from the feature extraction method.
2. Calculate the precision, recall and F1-score for each model.
3. Compare results and determine whether DTW is overall a good detection method and which feature extraction method is best suited.

1.4 Scope

What you didn't consider.

1.5 Layout of report

* Chapter 2 provides anu background information needed to understand how to solve the given problem as well as any helpful literature of previous solutions to the problem.
* Chapter 3 provides an explanation of the overall setup and method used to solve the problem including what metrics were used to meet the stated objectives.
* Chapter 4 gives a detailed design explanation for all the functional blocks defined in the previous chapter.
* Chapter 5 provides evidence of results obtained after the system has been designed as well as descriptions of what the results show.
* Chapter 6 contains all concluding thoughts on the results and what that means for the solutions as well as improvements and recommendations for the project.

My funky thesis (MFT) is as good as MFT can be, as Table 1.1 and Figure **??** shows. Have a look at my explanation in Section **??** on page **??**, which is in Chapter 2.

Table 1.1: Your table caption

	Centered number [Units]	R-aligned number	Width-set col
Description	100	140	24601

Chapter 2

Literature

2.1 Dynamic Time Warping

DTW is an algorithm for detecting the similarity of two signals, $X = [x_1, x_2, \ldots, x_n]$ and $Y = [y_1, y_2, \ldots, y_m]$. This measure of similarity is achieved by calculating an optimal alignment cost between the two signals of length $N \in \mathbb{N}$ and $M \in \mathbb{N}$ respectively. In terms of feature vectors such as STFT or MFCC this means that each frame making up one signal is compared to each frame making up another signal. Using an alignment cost the best alignment for each frame in one to the other signal is chosen and an overall DTW cost is calculated to show how similar the overall signals are. Figure 2.1 shows that such an optimal alignment path may mean that a single frame in signal X may produce the lowest cost against more than one frame in signal Y.[7][8]

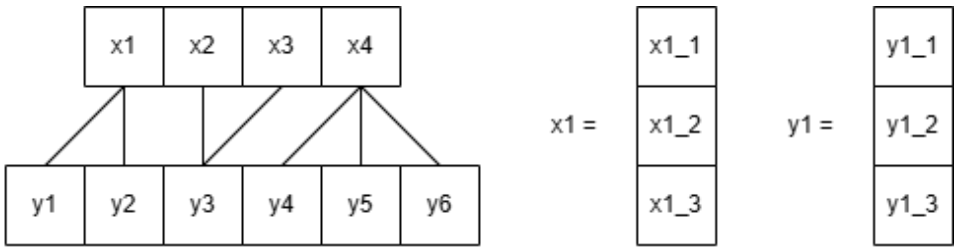


Figure 2.1: An example of the optimal warping path between two vector features, X with length N = 4 and Y with length M = 6

The overall cost needed to determine how similar the two signals are is calculated via a cost matrix. Cost matrix, $C_{N+1,M+1}$ for signals X and Y, is initialised such that $C_{i,0} = \infty$ for i = 1 to M, $C_{0,j} = \infty$ for j = 1 to M and $C_{0,0} = 0$.

The formula used to fill each cell of the cost matrix is:

$$C_{i,j} = \text{dist}(x_i, y_j) + \text{Q} \qquad \text{with} \quad \text{Q} = \min \begin{cases} C_{i-1,j-1} & \text{(match)} \\ C_{i-1,j} & \text{(insertion)} \\ C_{i,j-1} & \text{(deletion)} \end{cases} \tag{2.1}$$

$dist(x_i, y_j)$ represents the difference between X_i and Y_j as a distance. For vector signals this distance is calculated between each frame i of signal X and each fram j of signal Y. This can be done using various different techniques including cosine, euclidean or dot product similarity. For this project only the cosine similarity and euclidean distance metrics were used.[8]

Euclidean distance is sensitive to the magnitude and direction of the vectors for which it is calculated. In basic terms it is the absolute difference between two vectors.[9]

$$d(\mathbf{x_i}, \mathbf{y_j}) = \sqrt{(\mathbf{x_{i1}} - \mathbf{y_{j_1}})^2 + (\mathbf{x_{i2}} - \mathbf{y_{j_2}})^2 + \cdots + (\mathbf{x_{i_n}} - \mathbf{y_{j_n}})^2} \tag{2.2}$$

Cosine similarity is only sensitive to the direction of the vectors for which it is calculated. In basic terms it is the angle between the two vectors.[9]

$$\cos(\mathbf{x_i}, \mathbf{y_j}) = \frac{\mathbf{x_i} \cdot \mathbf{y_j}}{\|\mathbf{x_i}\| \, \|\mathbf{y_j}\|} \tag{2.3}$$

Q is the minimum value of three possible options namely match, insertion or deletion. Match is the value in the matrix cell diagonally down and left from the current cell for which the cost is being calculated. Insertion is the value in the matrix cell directly below the current cell and deletion is the matrix cell directly to the left of the current cell.

The cost which is then calculated for the current cell $C_{i,j}$ is a summation of the distance between the feature frames i and j in signals X and Y respectively as well as the lowest accumlated cost up until that point. This means that the final cell reached within the cost matrix, $C_{N+1,M+1}$ will contain the lowest overall cost for the similarity of signals X and Y. The lower the final cost, the higher the similarity between the two signals. The example shown in Figure 2.2 is shown if X and Y are chosen to be scalar signals, however the same logic is used for when X and Y are vectors.[8]

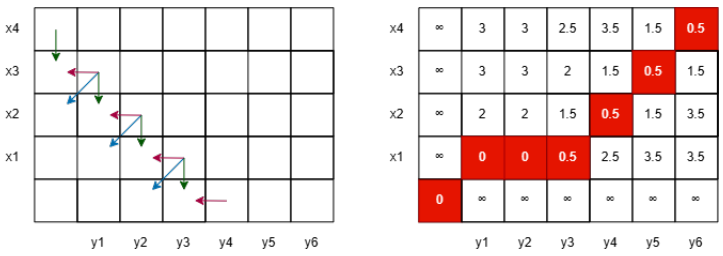


Figure 2.2: An example for the three possible options for min as well as an example of the optimal warp path found for signals X and Y assuming X=[0, 2, 1, 0] and Y=[0, 0, 0.5, 2, 1, 0]

The general case of DTW finds global alignment between the two signals. This alignment is found using the warping path. The warping path is the path of the best alignment between signals X and Y within the cost matrix, as shown in Figure 2.2. This path, W, is a sequence of matrix co-ordinates, (i, j), with a total length k.[10]

The boundary condition for global alignment ensures the final path starts and ends with the start and endframes of each respective signal.

$$i_1 = 1, i_k = N \quad \text{and} \quad j_1 = 1, j_k = M \tag{2.4}$$

The monotonicity condition ensures that each new point in the warp path is a step forward in time from the previous step.

$$i_{t-1} \leq i_t \quad \text{and} \quad j_{t-1} \leq j_t \tag{2.5}$$

The continuity condition ensures the step size of the warping path is only one unit in time.

$$i_t - i_{t-1} \leq 1 \quad \text{and} \quad j_t - j_{t-1} \leq 1 \tag{2.6}$$

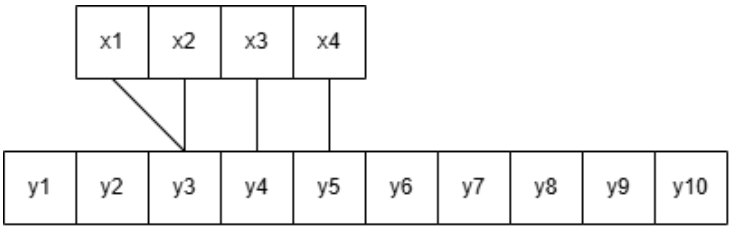


Figure 2.3: Example of a subsequence DTW alignment vetween two signal $X = [x_1, x_2, x_3, x_4]$ and $Y = [y_1.y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9, y_10]$

Subsequence DTW is a version of the DTW algorithm that alters the boundary condition such that global alignment is no longer necessary, instead the entirety of signal X is now matched with a subsequence of signal Y. The new boundary condition allows the warping path to start and end at anypoint within the first and final row of the matrix respectively.

$$i_1 = 1, i_k = N \quad \text{and} \quad 1 \leq j_1 \leq j_k \leq M \tag{2.7}$$

In global alignment the final cell in the cost matrix (N+1, M+1) is considered the overall optimal cost for those two signals. With subsequence DTW the entire final row of the matrix is checked to find the lowest cost. The position of that cost shows where the end point of the alignment detected via subsequence DTW is.[11]

2.2 Feature Extraction Methods

2.2.1 Short Time Fourier Transform

The STFT is used to analyse the change in frequency of a signal over time. The STFT of a segment of audio is calculated by multiplying the signal in the time domain, $x(t)$ with a window $w(t-\tau)$ which is sweeped across said signal. The Fourier Transform of each window is calculated and used to determine the frequency information for that window. Thus, the STFT provides the change in frequency for a segment of audio throughout time.[12][13] The mathematical formulation of STFT is:

$$X(\tau,\omega) = \int_{-\infty}^{\infty} x(t)\, w(t-\tau)\, e^{-j\omega t}\, dt, \tag{2.8}$$

where τ is the position of the window as it shifts through time at step-size H and ω is the frequency being analysed.

The frequency resolution of the STFT is given as:

$$resolution = \frac{f_s}{L} \tag{2.9}$$

where f_s is the sampling frequency of the signal and L is the length of the window overwhich the FT is taken.

This means that the STFT has a trade-off between time and frequency resolution.

If L is kept short, time resolution is high since smaller segments of time are being used for one FT analysis. However, frequency resolution is then decreased since too little time passes for the FT to distinguish between frequencies spaced closer together.

If L is kept long, time resolution drops. This means that fast changes in frequency will be missed. However, frequency resolution increases and it becomes easier to distinguish between differrent frequency bands.

Window overlap is another consideration when using STFT. Increased overlap creates a smoother representation of the changes in frequency occuring over time since information at the edges of the window is no longer lost. The trade-off with overlap is the increase in computational complexity that will occur due to smaller and thus more windows of a signal being analysed.[7][13]

2.2.2 Mel-Frequency Cepstral Coefficients

MFCC is a common choice for feature extraction when working on aucoustic data. This is because it is set up to percieve frequencies as they are percieved by the human ear. That means the lower frequencies are given a higher resolution than the high frequencies.[14] Features are initially extracted using the FFT. After this they are passed through a

Mel-scale Filter bank. The formula for the mel-frequency calculation is:

$$Mel(f) = 1127 \ln\left(1 + \frac{f}{700}\right) \tag{2.10}$$

with $Mel(f)$ as the frequency value on the mel scale and f as the actual frequency in Hz.[12]

This bank is made up 20 to 40 triangular filters which are placed ,smallest width to largest width according to the mel scale, along the frequency spectrum. The standard amount of triangular filters chosen is 26. This results in 26 coefficients for the entire range of frequencies being analysed.[15][14]

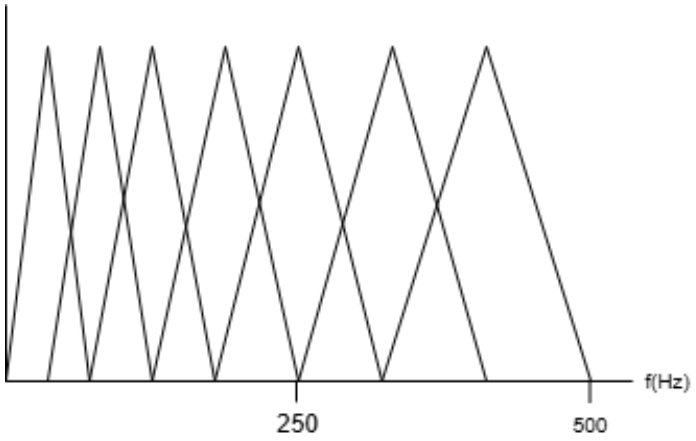


Figure 2.4: The mel scale filter bank is made up of 20 to 40 triangular filters arranged according to the mel scale.

The log of these coefficients is then taken and finally the Discrete cosine transform (DCT) is used to decorrelate the filter bank coefficients. The final number of cepstral coefficients returned by the DCT is typically 12 or 13 since the most important information is kept there. The formula use to perform the DCT is:

$$MFCC_i = \sum_{k=1}^{K} X_k \cos\left(\frac{i(k-0.5)\pi}{K}\right), \quad i = 1, 2, \ldots, K \tag{2.11}$$

with i as the index of the current coefficient, K as the total number of triangular filters used, k as the specific filter bank at that point in time and X_k as the energy after log was applied to the mel coefficients.[12]

The first and second derivatives, often referred to as deltas, of the cepstral coefficients are often used in feature extraction. They show the change in frequency between cepstral coefficients. This can be vital information when analysing how frequency changes across time since the MFCC process tends to lose this information. The formula for the derivatives is:

$$x_t^{(d)} = \frac{\sum_{m=-M}^{M} m \cdot x_{t+m}^{(d-1)}}{\sum_{m=-M}^{M} m^2} \tag{2.12}$$

with t as the specific point in time, d as the order of derivative and m as the frame offset.[16][17][18]

2.2.3 Linear Predictive Coding

Chapter 3

Dataset

3.1 Background

The dataset used contains three 72 hour recordings, recorded in False Bay, near Gordons Bay, between 22 April 2023 and 22 June 2023. The recordings were taken at a depth of 20 to 30 meters at the coordinates, S 34.16813deg, E18.758718deg.

The device used for the recordings was a single omnidirectional hydrophone, designed and built by a Stellenbosch University research group from the Electrical and Electronic Engineering Department.

The recorded data has a sampling rate of 8kHz and a 16-bit resolution. The data was extracted as a 72 hour wav file from an SD card used by the recording device.[19]

3.2 Labeling

Detection tables, containg start and end times of calls within the each file, were provided from a previous research project which used a band-limited energy detector to flag calls. This method unfortunately did not detect all the calls within the audio files and none of the calls were labelled as a specific type of call.

Another detection table was procured. This table contained more detections for the first 72 hour file and contained call labels for all three files. This meant that there were sufficient labels for the first 72 hour recording of the dataset and there were partial labels for the second and third 72 hour recordings within the dataset.[19]

3.3 Data Splitting

For this project the data was plit in three sections. Each of the 72 hour audio files was chosen for a different section. Given that only the first of the three 72 hour audio files was

well labeled, it was decided that this would be used for training and testing. The second audio file was used for hyperparameter tuning and the third was used to select templates.

3.3.1 Template Selection

The audio file which was recorded starting on the 29th was used to select templates. These chosen templates were used for the comparison which determined whether an audio clip contained calls or not.

For the three different call type, single-tones, multi-tones and burst-tonals, the maximum number of templates per call type which could be extracted such that each call type had the same amount of templates was 10. The limiting call type was burst-tonals.

To ensure that the 10 templates chosen were representative of varying amounts of background noise templates were chosen by looking at spectrograms of detected calls throughout the audio file. The visual inspection was also necessary since some of the calls labeled as single-tones looked more like multi-tones and some of the multi-tones could actually be classified as burst-tonals.

3.3.2 Hyperparameter Tuning Template Selection

The audio file which was recorded starting on the 26th was used to select calibration templates. These templates were used to determine below which threshold a segment of audio could be considered a call. For the calibration templates 10 templates were chosen for each call type. This was again limited by the burst-tonals maximum number of templates. In this file there were however only 9 labeled burst-tonals, but when selecting multi-tones it was found that a call originally labeled as a multi-tone could in fact be seen as a burst-tonal.

Besides these 30 templates, 10 for each call type, 100 background templates were also selected. Within each audio file most of the time is made up of background noise and only some parts actually contain calls. This was simulated by selecting many more background templates that call-type templates. Since there is no table labeling background noise background segments of varying sizes were extracted from the file and 100 templates were manually selected. The reason for the manual process links with the unreliability of the labelled data. The audio file used was only labeled by the band-limited energy detector and thus did not label all the calls within the file. This means that some of the samples extracted as background audio actually contained calls that had never been labeled. Manually selecting each template ensured that all the templates chosen were in fact background noise only.

3.3.3 Evaluation Data Selection

The entire first audio file which was recorded starting on the 22nd was used for the final evaluation of the whale call detection model. No templates were extracted, instead the entire file was sweeped to search for calls.

The 22nd file was the only file which contained tonal-downsweep calls. Due to the limited amount of these types of calls all the detected tonal-downsweeps within this file was ignored while testing since thresholds could not be trained for this call type.

Chapter 4

System Design

4.1 Overview

The project was coded in Python. The functionality of each section of the project was placed in relevant .py files which were imported to the relevant jupyter notebook which was used for that section.

4.2 Feature Extraction Methods

The raw audio recorded by the hydrophone was stored in .wav files with a sampling frequency of 8kHz. Before extracting feature vectors from audio segments the audio was first downsampled to 1kHz sampling rate. This helped remove unecessary frequency information as the Nyquist Frequncy ($f_s/2$) for a 1kHz sampling rate is 500 Hz.[20]

Most of the Bryde's Whale vocalizations occur below 200 Hz. Thus, this sampling rate also ensures no relevant information was lost before analysis.[2]

4.2.1 STFT

The signal.stft function from the scipy library was used to calculate the STFT magnitudes for use as feature vectors. Window length L, as referenced in Eq.(2.9), is calculated by multiplying f_s, the sampling frequency of the given audio segment, with the chosen STFT frame duration. No set frame duration was selected as the optimal frame duration was chosen during the calibration stage of this project.

The window ovelrap H, as referenced in Eq.(2.9), is calculed by multiplying L with the chosen overlap percentage. For this project the overlap percentage was chosen as 0,75. The high overlap percentage produced a smooth spectrogram when viewing the time–frequency representation of the given audio segment. A Hamming window was chosen to decrease

any spectral leakage caused by the finite length of the windows used to extract the feature vectors.[13][21]

4.2.2 MFCC

The feature.melspectrogram and feature.mfcc functions from the librosa library were used to calculate the cepstral coefficients for the MFCC feature extraction implemetnation. The window length for the Fourier Transform required is calculated by multiplying f_s with the chosen frame duration which is optimally selected during calibration. Unlike the STFT function librosa.feature.mfcc requires the hop length and not the ovrelap length. This is calculated by multiplying $L - H$. H is also calculated by choosing the overlap percentage to be 0,75. Just as in the STFT feature extraction implementation a Hamming Window is used to decrease spectral leakage.

The librosa function used for calculating MFCC automatically uses DCT type 2 and ortho normalization. These settings mean that the bottom row of the output matrix, c_0, represents the sum of the log-spectrum energy. It is common practice in speech analysis to remove coefficient 0, as it contains total energy and can dominate subsequent analysis. For this project, only coefficients 1 to 14 were kept. This ensured that the analysis considered only the overall shape of the log spectrum and not the additive energy. The number of n_{mels} was chosen as 26, which is common practice in audio processing, and 14 coefficients were calculated because 13 are typically used, with the lowest coefficient 0 removed.[22][23][24]

4.2.3 LPC

4.3 DTW methodology

The sequence.dtw function from the librosa library was used to calculated the Dynamic Time Warping cost matix between a template and an audio segment. The distance matrix needed as input for the function was calculated using the spacial.distance.cdist function from the scipy library.

The distance function was given the template feature and the audio segent feature extracted either by STFT or MFCC as well as the metric specified for each extractor. The two metrics considered were cosine similarity and euclidean distance. As shown in Eq.(2.3) and Eq.(2.2), cosine similarity calculates the distance in angle between two points and eiclidean distance calculates the acual distance between the two point. For the STFT feature extraction method cosine similarity was used rather than euclidean distance since cosine does not depend on magnitude an thus the distance between a template and a call further away vs closer together will only be judged on the angular difference and not the difference in magnitude. This ensures that louder and softer calls of the same type can be classified by spectral shape only and not loudness. For the MFCC feature extraction

implementation the energy created by the call due to how close or far away it is from the recording device is removed when c_0 was removed as explained in Section4.2.2. This means that euclidean distance could be used without concern of the effect of magnitude on the distance since only spectral shape is left to be analysed. Euclidean distance is the common choice for DTW analysis, therefore since the main cosine similarity benefit is not necessary for MFCC Euclidean distance was chosen.[25]

The subsew setting with in the sequence.dtw function was set to True. As explaines in Section 2.1, subsequence DTW means that the start and end points of a detected match do not have to be the first and last points of both signals. When searching a 72 hour audio file for whale calls subsequence DTW allows for the detection of multiple calls within a single audio segment by searching the final row of the cost matrix for dips and choosing the lowest point of each dip as an end point of a call. This is further explained in Section4.5.[26]

4.4 Hyperparameter Tuning

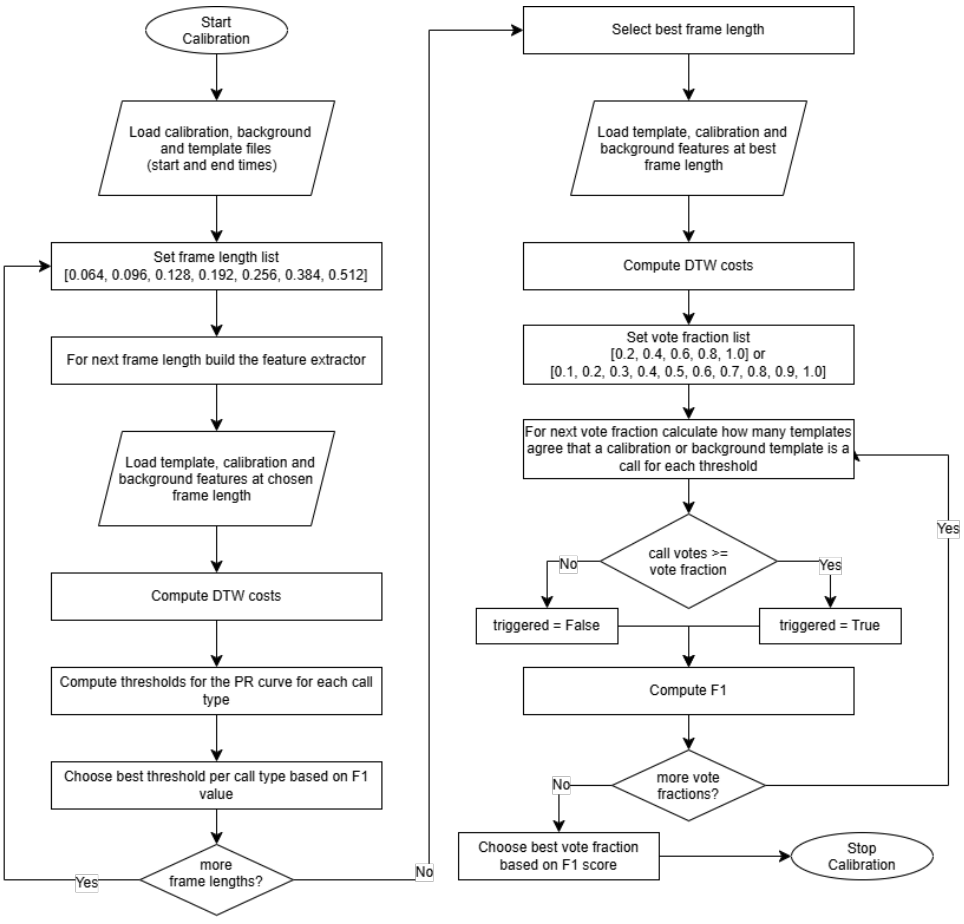


Figure 4.1: This is a software flow diagram describing how the hyperparameters are chosen for the project.

The frame length of any feature extraction method affects the frequency resolution. Therefore, different widths of frame length (0.048, 0.064, 0.096, 0.128, 0.192, 0.256, 0.384, 0.512) were tested to see which produced the best F1 score. In the process of determining the best frame length the F1 score is calculated by also finding the best possible thresholds for which each call can be seperated as either a call or background noise. While the project only focused on general detection of calls and not on classification, three different thresholds, one for each call type, were calculated and used. The reason for this is that choosing one overarching threshold would make the model far too lenient for some calls as call types such as single-tones and burst-tonals tend to have a significant difference in threshold choice. Even though three different thresholds were chosen based on the templates from each call type in the final detection only general call detection was considered and not whether the model classified the calls correctly. This was due to the fact that single0tones and multi-tones sometimes look very similar and thus tend to be incorrectly classified.

When deciding whether an audio segment is a call or not that segment is compared against all 30 templates (10 for each call type). For each call type some of the 10 templates will say that it is a call and others will say that it is not since the templates were chosen for varying noise conditions. The vote fraction is used to decide whether the audio segment is from that call type or not. If that vote fraction amount of the 10 calls say it is a call then it will be classified as such if less than that vote fraction agree that it is a call then it will not be classified as a call. The F1 score is calculated using different voting fractions (0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0) as well as the chosen frame length and thresholds. This F1 is used to choose the best possible voting fraction for all three call types.

4.5 Training Procedure

4.6 Evaluation Metrics

Chapter 5

Results

5.1 Experiment 1: STFT Features

5.1.1 5 Templates

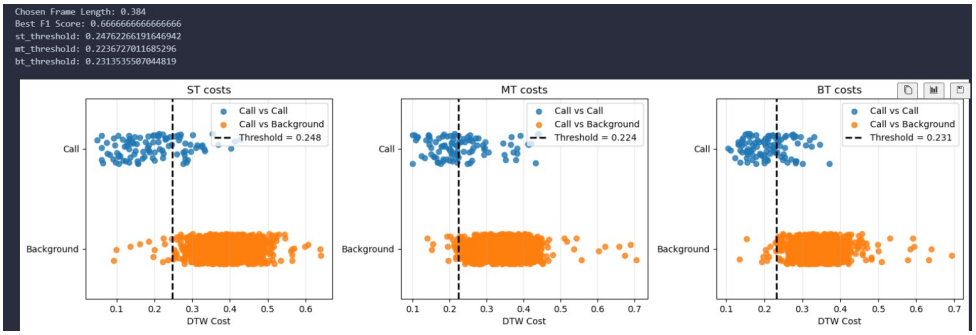


Figure 5.1: Scatter plot showing threshold placement in terms of background images and call templates

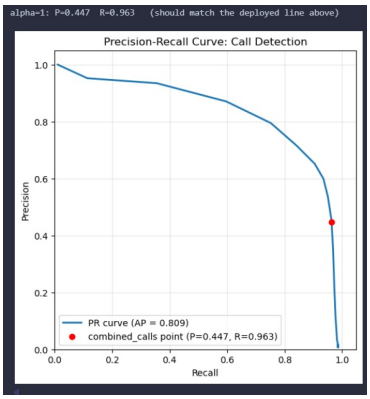


Figure 5.2: Precision vs Recall curve

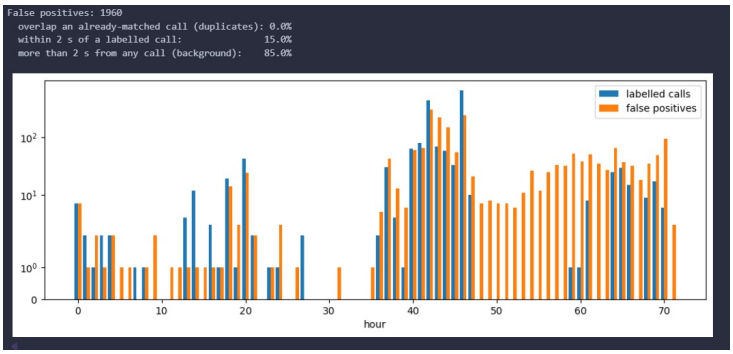


Figure 5.3: Timeline of audio file 22 showing where the most False Positives occur in relation to the actual calls

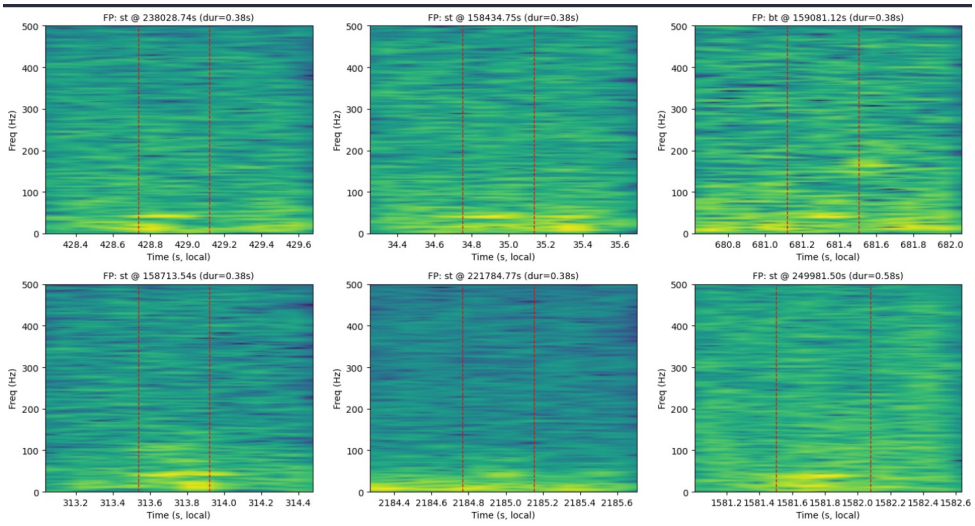


Figure 5.4: Spectrograms of False Positive calls. Many of which could actually be real calls

5.1.2 10 Templates

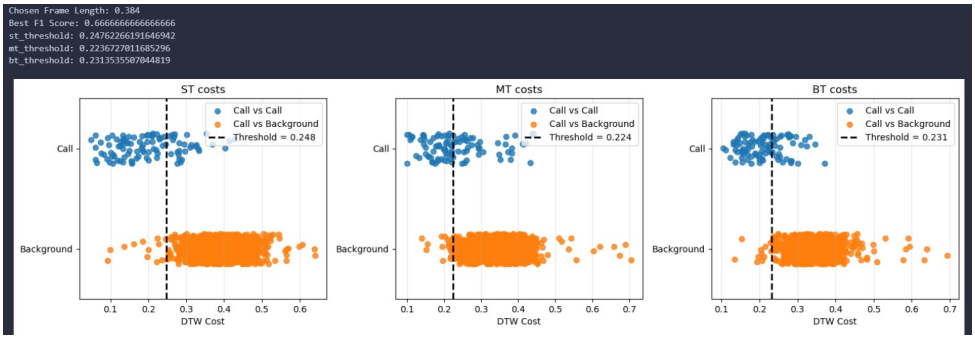


Figure 5.5: Scatter plot showing threshold placement in terms of background images and call templates

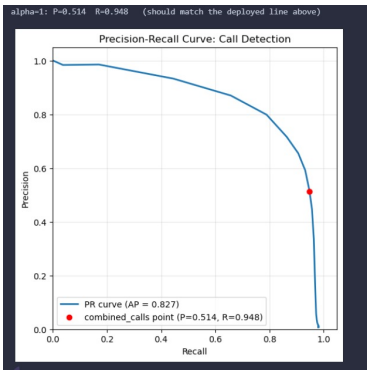


Figure 5.6: Precision vs Recall curve

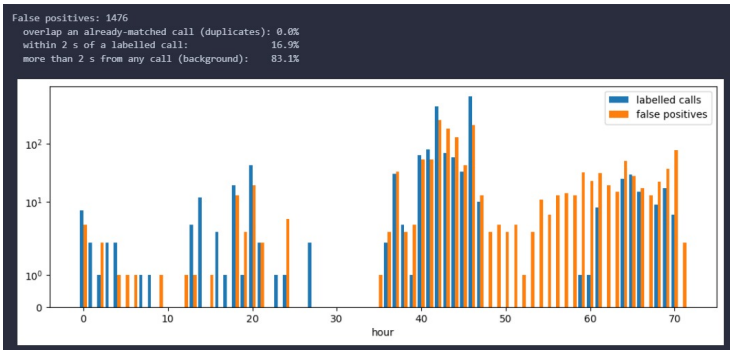


Figure 5.7: Timeline of audio file 22 showing where the most False Positives occur in relation to the actual calls

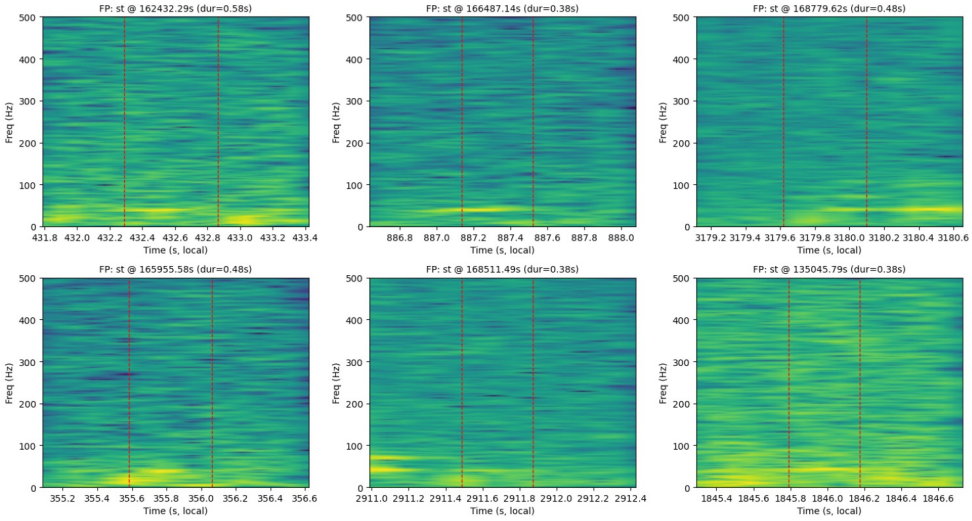


Figure 5.8: Spectrograms of False Positive calls. Many of which could actually be real calls

5.2 Experiment 2: MFCC Features

5.2.1 5 Templates

5.2.1.1 No derivatives

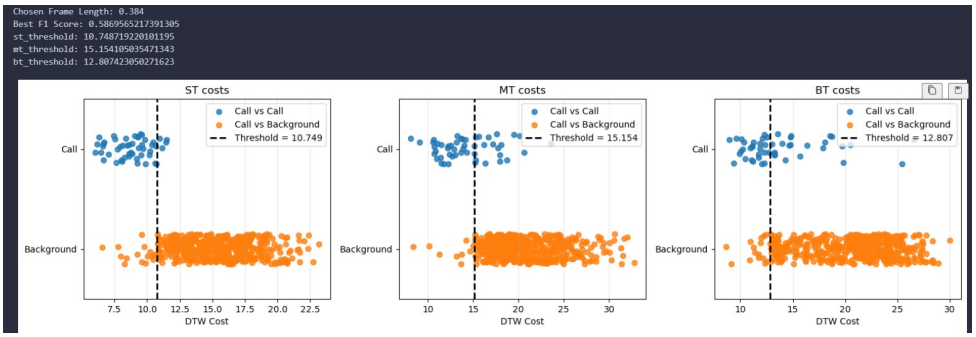


Figure 5.9: Scatter plot showing threshold placement in terms of background images and call templates

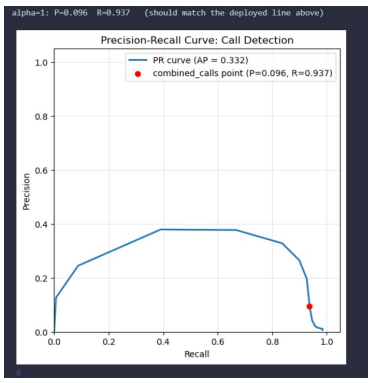


Figure 5.10: Precision vs Recall curve

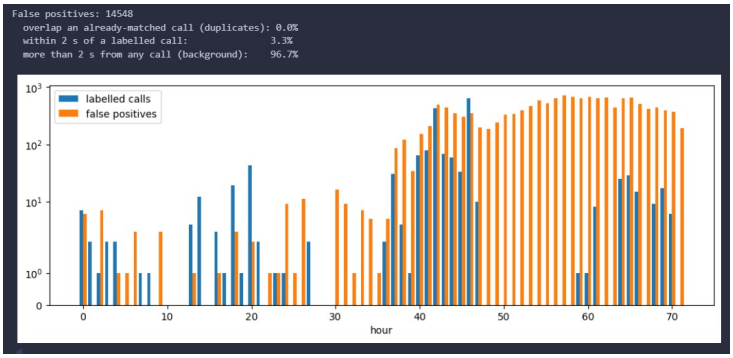


Figure 5.11: Timeline of audio file 22 showing where the most False Positives occur in relation to the actual calls

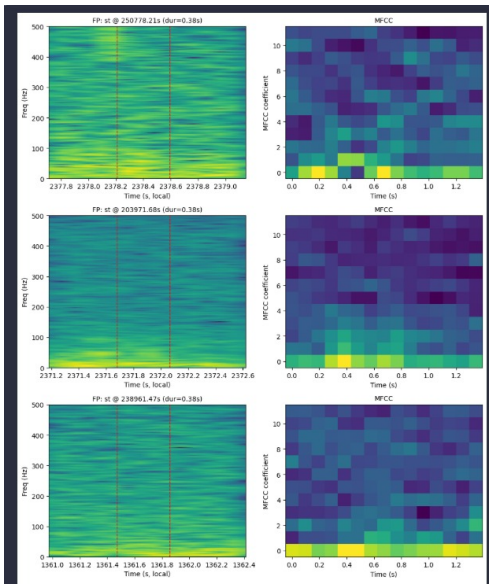


Figure 5.12: Spectrograms of False Positive calls. Many of which could actually be real calls

5.2.1.2 First derivative

5.2.1.3 First and second derivatives

5.2.2 10 Templates

5.2.2.1 No derivatives

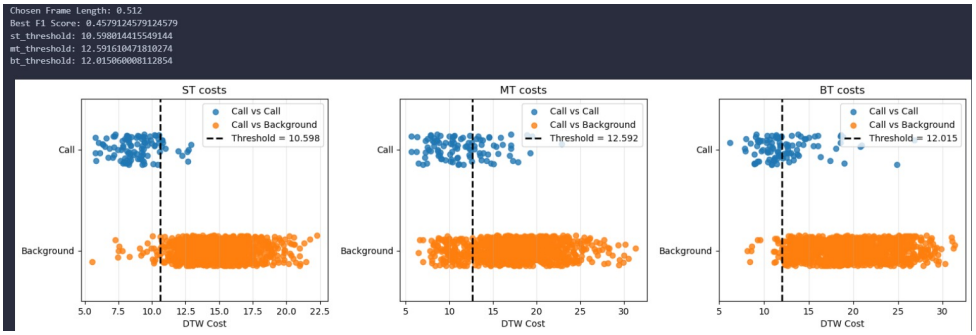


Figure 5.13: Scatter plot showing threshold placement in terms of background images and call templates

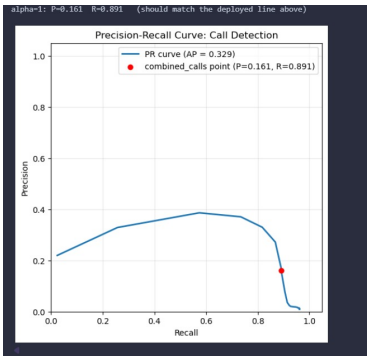


Figure 5.14: Precision vs Recall curve

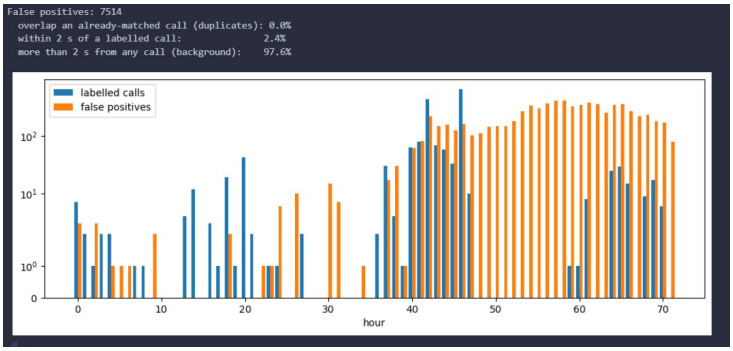


Figure 5.15: Timeline of audio file 22 showing where the most False Positives occur in relation to the actual calls

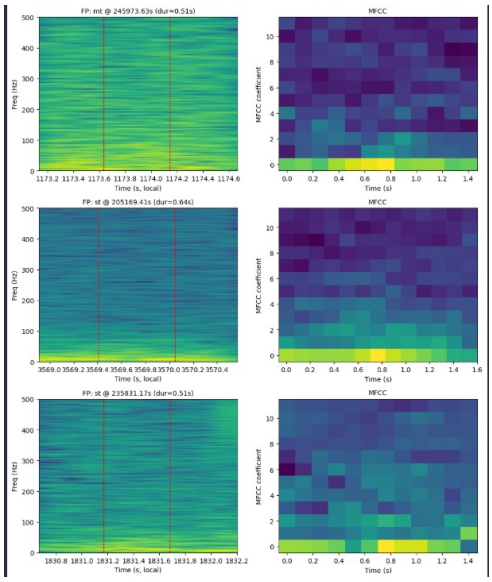


Figure 5.16: Spectrograms of False Positive calls. Many of which could actually be real calls

5.2.2.2 First derivative

5.2.2.3 First and second derivatives

5.3 Experiment 3: LPC Features

5.3.1 5 Templates

5.3.2 10 Templates

Chapter 6

Conclusion

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Appendix A

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